

Problem Statement

Region-Aware RAG to Close LLMs' Cultural Gap (BLEnD)

1) Group Name : Team BlenD

2) Team Name, Roll no :

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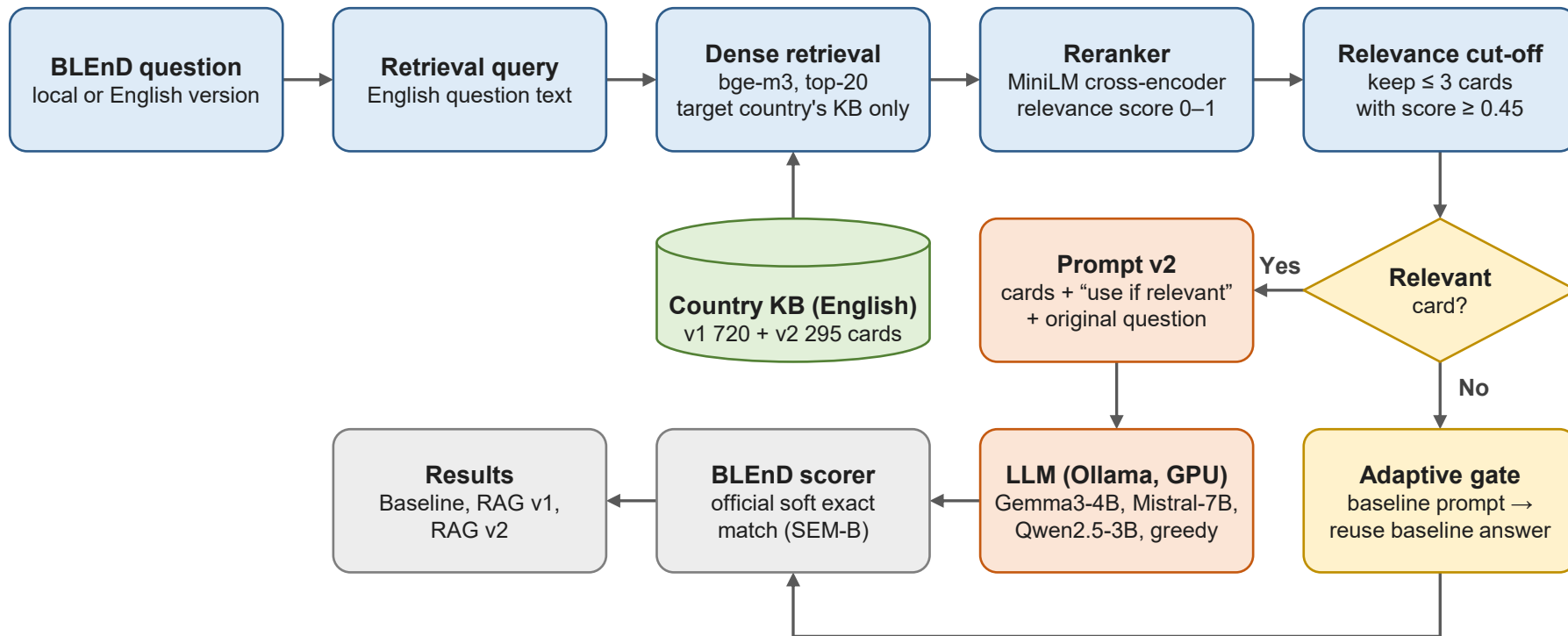
3) Novelty :

Cultural Knowledge Base per Country + Hybrid Adaptive RAG that improved LLM performance on Low resource Countries. Low-resource
Accuracy 28.3% → 48.7%;

4) Contributions :

- Member 1 - Siddharth Rai: Knowledge Base, RAG v2.
- Member 2 - Gaurav Kumar: Baseline, RAG v1

Architecture



Results and Findings

Country	Lang.	Baseline	RAG v1	RAG v2	Δ v2
US	en	74.9	53.5	73.2	-1.7
Spain	es	56.3	36.8	50.0	-6.3
	en	50.8	34.3	44.5	-6.4
South Korea	ko	48.0	39.2	43.7	-4.3
	en	47.7	35.8	42.3	-5.4
Azerbaijan	az	23.6	18.0	53.3	+29.7
	en	47.6	30.6	65.4	+17.9
Ethiopia	am	7.9	12.3	28.0	+20.1
	en	34.1	23.7	48.0	+13.9
All 9 settings		43.4	31.6	49.8	+6.4
High-resource avg		55.5	39.9	50.7	-4.8
Low-resource avg		28.3	21.2	48.7	+20.4

Key findings
RAG v1 (always top-3, KB v1) lowered accuracy almost everywhere: -11.9 points on average
Cause: the KB held the answer for only 7–14% of questions; models refused or copied wrong cards
RAG v2 (question-driven KB + rerank + cut-off $\tau = 0.45$ + gate) improves every AZ/ET setting for all 3 models (+8.6 to +36.9)
High–low resource gap: 27.2 $\rightarrow$ 18.7 (v1) $\rightarrow$ 2.0 (v2) points
Cost: high-resource countries (old KB only) lose 1.7–6.4 points
SEM-B %, mean of 3 models $\times$ 2 prompts (official BLENd scorer). High-resource = US, ES, KR (α , τ tuned here); low-resource = AZ, ET (held out).